



Modelling Financial Markets

Chapter 12: Options and the Volatility Surface



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ATM: At-The-Money
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CRR: Cox–Ross–Rubinstein (binomial tree)
- ▣ DVOL: Deribit Volatility Index
- ▣ ETF: Exchange-Traded Fund
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ ITM: In-The-Money
- ▣ MA: Moving Average
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ OLS: Ordinary Least Squares
- ▣ OTM: Out-of-The-Money
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ RV: Realised Variance
- ▣ SSVI: Surface SVI (Gatheral–Jacquier parameterisation of the whole surface)
- ▣ SVI: Stochastic Volatility Inspired (parameterisation)
- ▣ SVIX: Simple VIX: index of simple variance swaps (Martin, 2017)
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index

Acronyms used in this material (2/2)

- VIX3M: Cboe 3-Month Volatility Index
- VIX9D: Cboe 9-Day Volatility Index
- VRP: Variance Risk Premium
- VVIX: Cboe VIX of VIX Index
- XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Motivation

Parity and Static Replication

The Binomial Tree

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Bitcoin Options and the Volatility Surface

The VIX and the Variance Risk Premium

Options with Zero Days to Expiry

Case Study: Are Option-Implied Bounds Valid and Tight?

AI for Scientific Discovery

Summary

References

Why Options?

What an option does

- An **option** gives the right, not the obligation, to buy or sell an asset at a fixed price
- It pays only in some states of the world: a bet on the **distribution**, not only on the direction
- Option prices reveal what the market expects about volatility and crashes

What we do in this chapter

- Derive prices by replication: the Black–Scholes equation and the static spanning of any pay-off by options
- Derive what a hedge leaves behind: discrete-hedging error and the gamma-weighted volatility P&L
- Estimate the smile and the risk-neutral density under no-arbitrage constraints, with confidence bands
- Test the VIX as a forecast and the variance premium as a predictor, with the right standard errors

Chicago, 1973: Listed Options Begin



Trading floor of the Chicago Board of Trade in 2006; its members founded the Cboe in 1973

Photo: [Richie Diesterheft \(2006\)](#); CC BY 2.0; [Wikimedia](#)

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- The Chicago Board Options Exchange (Cboe) opened in 1973 as the first exchange for standardised, listed options [MacKenzie & Millo \(2003\)](#)
 - ▶ Before: options were negotiated one by one, over the counter
- The same year: the Black–Scholes and Merton pricing papers [Black & Scholes \(1973\)](#), [Merton \(1973\)](#)
- Traders soon used the formula on the floor: theory changed the market it described [MacKenzie & Millo \(2003\)](#)
- Textbooks: [Franke, Härdle & Hafner \(2019\)](#), [Härdle, Chen & Overbeck \(2017\)](#), [Gatheral \(2006\)](#)

The Formula and Its Authors



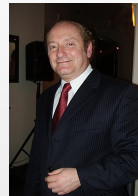
Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia;
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Myron Scholes (2008)

Photo: Nobellaureatesphotographer (2008);
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Robert C. Merton (2006)

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- Key idea: a portfolio of the stock and cash can **replicate** the option, so its price follows from no-arbitrage [Black & Scholes \(1973\)](#)
- Scholes and Merton received the [1997 Nobel Memorial Prize in Economic Sciences](#); Black had died in 1995

Data Used in This Chapter

Series	What it measures	Period	Owner
S&P 500	US equity index, daily close	1990–2026	S&P Dow Jones Indices
SPY	ETF on the S&P 500: open and close	1993–2026	State Street
VIX	30-day implied volatility of the S&P 500	1990–2026	Cboe
VIX9D, VIX3M	the same over 9 days and 3 months	2011 / 2007–2026	Cboe
VVIX	implied volatility of the VIX	2006–2026	Cboe
Bitcoin options	all listed contracts, one dated snapshot	22 Sep 2026	Deribit
DVOL	30-day implied volatility of Bitcoin	2021–2026	Deribit

- Daily data up to 18 September 2026; the option snapshot was taken on 22 September 2026, 07:51 UTC
- Implied volatility is quoted in annualised %; “vol points” = percentage points of volatility

Notation and Prerequisites

- European call and put pay off $(S_T - K)^+$ and $(K - S_T)^+$ at expiry T ; $\tau = T - t$ in years; American options can be exercised at any time up to T
 - ▶ Both pay-offs are convex in S_T : the value of this convexity is what the smile prices
- Forward $F = Se^{(r-q)\tau}$, log-moneyness $k = \ln(K/F)$; ATM: $k \approx 0$; OTM: puts with $K < F$ and calls with $K > F$; ITM: the opposite
 - ▶ Most of this chapter works in k and in total implied variance $w = \sigma_{imp}^2 \tau$
- No-arbitrage bounds for European options on a non-dividend-paying asset, without any model:
 $\max(S - Ke^{-r\tau}, 0) \leq C \leq S$ and $\max(Ke^{-r\tau} - S, 0) \leq P \leq Ke^{-r\tau}$
 - ▶ So, with $r > 0$ and $\tau > 0$, an American call on a stock without dividends is never exercised early: it is worth at least $S - Ke^{-r\tau} > S - K$; with $r < 0$ early exercise can be optimal
- Assumed: Brownian motion, Itô's lemma and martingales (Chapter 11); regression with HAC standard errors (Chapter 8)

Put–Call Parity

- Two portfolios with the same pay-off at T (European options, same K and T , no dividends):
 - ▶ A: one call + cash Ke^{-rT} (worth K at T) $\Rightarrow \max(S_T, K)$
 - ▶ B: one put + one share $\Rightarrow \max(K, S_T)$
- Same pay-off in every state, so the same price today, otherwise there is an arbitrage:
 - ▶ $C_t - P_t = S_t - Ke^{-rT}$ (with dividend yield q : S_te^{-qT} instead of S_t)
- No model for S is needed: parity uses only no-arbitrage [Merton \(1973\)](#)
- Consequence: a call and a put with the same K and T have the **same** implied volatility (Section 6)

Static Replication: the Carr–Madan Spanning Formula

- Any twice-differentiable pay-off $f(S_T)$ and any point F (Taylor with integral remainder, split at F):
 - ▶
$$f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+ dK + \int_F^\infty f''(K)(S_T - K)^+ dK$$
- Proof for $S_T > F$: $\int_F^{S_T} f''(K)(S_T - K) dK = f(S_T) - f(F) - f'(F)(S_T - F)$ by parts; the put integral vanishes; symmetric for $S_T < F$
- Take $e^{-r\tau} E^Q[\cdot]$ with F the forward (a forward contract costs 0):
 - ▶
$$V_0 = e^{-r\tau} f(F) + \int_0^F f''(K) P(K) dK + \int_F^\infty f''(K) C(K) dK$$
- A bond, a forward and a strip of OTM options with weights $f''(K)$ replicate any smooth pay-off, whatever the model for S [Bakshi, Kapadia & Madan \(2003\)](#)

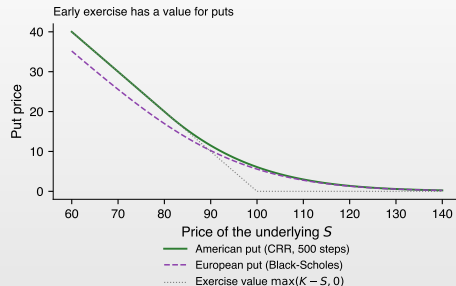
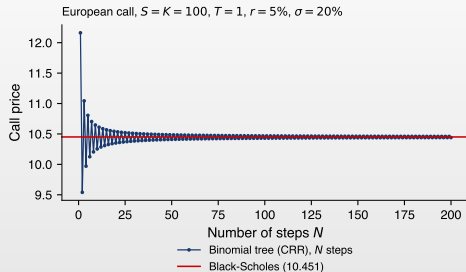
Three Corollaries of Spanning

- ▣ **Variance:** $f(S) = -2\ln(S/F)$ has $f''(K) = 2/K^2$: the strip with weights $1/K^2$ is the variance swap and the VIX (Section 8)
 - ▶ Low strikes get the largest weights, so an expensive put wing makes variance expensive
- ▣ **Density:** a butterfly $[C(K-h) - 2C(K) + C(K+h)]/h^2 \rightarrow \partial^2 C / \partial K^2 = e^{-r\tau} q(K)$ [Breedon & Litzenberger \(1978\)](#): the risk-neutral density is a price (Section 7)
- ▣ **Moments:** $f = [\ln(S/F)]^n$, $n = 2, 3, 4$, price the raw risk-neutral moments of the log return; centring and standardising them gives volatility, skewness and kurtosis [Bakshi, Kapadia & Madan \(2003\)](#); $f = S^2$ gives the SVIX of [Martin \(2017\)](#)
- ▣ Retail strategies (straddle, spread, protective put) are special cases: finite strips with f'' concentrated at one or two strikes

No Arbitrage and the Pricing Measure

- One period, $S \rightarrow uS$ or dS , $d < R = e^{r\Delta t} < u$: $\Delta = \frac{C_u - C_d}{(u - d)S}$ shares plus cash replicate the call, so
 - ▶ $C = \frac{q C_u + (1 - q) C_d}{R}$, $q = \frac{R - d}{u - d}$: q is the unique probability under which S_t/R^t is a martingale
- The real probability of an up move never enters: risk preferences are already in S
- **Fundamental theorem:** no arbitrage $\Leftrightarrow$ an equivalent martingale measure Q exists; complete market $\Leftrightarrow Q$ is unique [Harrison & Pliska \(1981\)](#), [Delbaen & Schachermayer \(1994\)](#)
 - ▶ With jumps or stochastic volatility the market is incomplete: many Q , and option prices carry risk premia (Section 8)
- CRR tree [Cox, Ross & Rubinstein \(1979\)](#): $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, value backwards from T ; American options: $\max(\text{continuation}, \text{exercise})$ at every node

From the Tree to Black-Scholes



□ Left: CRR price of a European call for $N = 1, \dots, 200$ steps; right: American vs European put

Reading the Binomial Results

- The tree converges to Black–Scholes (10.451) as N grows: $N = 50$: 10.411; $N = 200$: 10.441; $N = 1000$: 10.449
 - ▶ Convergence oscillates: $N = 10$ gives 10.253, $N = 11$ gives 10.611 (the strike falls on or between nodes)
- American put at $S = K = 100$: 6.090 vs European 5.574
 - ▶ Early-exercise premium 0.516: deep in the money it pays to receive K now and earn interest
- The tree is the workhorse for American options; Black–Scholes is its limit for European ones
- The error is of order $1/N$ and oscillates between even and odd N : $N \cdot$ error stays near -2.0 and $+1.75$ (Seminar 12, A4)

The Black–Scholes Model

- Price follows a **geometric Brownian motion**: $dS_t = \mu S_t dt + \sigma S_t dW_t$
 - ▶ W_t : Brownian motion, independent Normal increments $W_{t+h} - W_t \sim N(0, h)$
 - ▶ So log returns over τ follow the Normal distribution with variance $\sigma^2 \tau$: σ is the annualised volatility
- Assumptions: constant σ and r , continuous trading, no costs, short selling allowed, no arbitrage
- Continuous-time limit of the tree: the hedge Δ is adjusted at every instant [Black & Scholes \(1973\)](#), [Merton \(1973\)](#)
 - ▶ The drift μ drops out, exactly as the real probability did in the tree
- Chapter 11 derives these processes; here we use their consequences

Derivation: the Black–Scholes Equation

- ▣ Replicate $V(S, t)$ with a **self-financing** portfolio: Δ_t shares and cash B_t at r , $\Pi = \Delta S + B$; rebalancing is paid from the portfolio, so $d\Pi = \Delta dS + rB dt$ (no $S d\Delta$ term)
- ▣ Itô's lemma: $dV = (V_t + \mu SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}) dt + \sigma SV_S dW_t$
- ▣ Choose $\Delta = V_S$, start from $\Pi = V$ ($B = V - SV_S$): $d(V - \Pi) = (V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} - rB) dt$; both dW_t and μ drop out
- ▣ A riskless position that costs nothing must earn nothing: the drift is 0, hence the **Black–Scholes equation**
 - ▶ $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$, with $V(S, T) = \text{pay-off}$
- ▣ That μ is irrelevant is a result, not an assumption; Feynman–Kac: $V = e^{-r\tau} E^Q[\text{pay-off}]$ with S drifting at r under Q

The Black–Scholes Formula

- European call and put, no dividends (with yield q : replace S by $Se^{-q\tau}$):
 - ▶ $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \quad P = Ke^{-r\tau}\Phi(-d_2) - S\Phi(-d_1)$
 - ▶ $d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$ with Φ the standard Normal distribution function
- Reading the two terms:
 - ▶ $\Phi(d_2)$ = risk-neutral probability that the call ends in the money
 - ▶ $\Phi(d_1)$ = number of shares in the replicating portfolio (the delta)
- Five inputs: S, K, τ, r observed; σ is the only one not observed: it must be estimated or implied

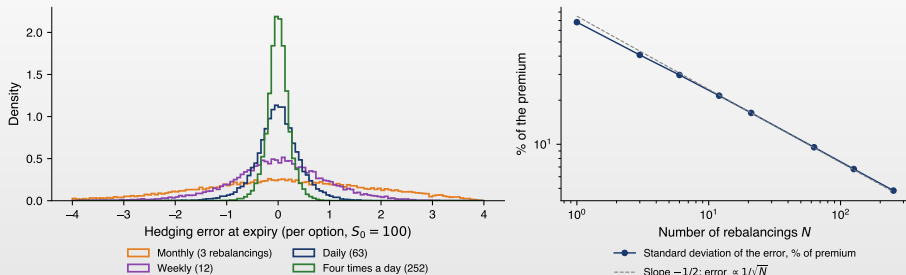
The Greeks: One Identity, All Sensitivities

- Delta $\Delta = \Phi(d_1)$ (put: $\Phi(d_1) - 1$); gamma $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$, with φ the standard Normal density
- Key identity: $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$, because $d_1^2 - d_2^2 = 2\ln(S/K) + 2r\tau$
- Vega: $\partial C/\partial\sigma = S\varphi(d_1)\partial_\sigma d_1 - Ke^{-r\tau}\varphi(d_2)\partial_\sigma d_2 = S\varphi(d_1)\partial_\sigma(d_1 - d_2) = S\varphi(d_1)\sqrt{\tau}$
 - Hence vega = $\Gamma S^2\sigma\tau$: for one expiry, gamma and vega are proportional (Seminar 12, A5–A6)
- Theta from the equation itself: $\Theta = rC - rS\Delta - \frac{1}{2}\sigma^2 S^2\Gamma$; rho = $K\tau e^{-r\tau}\Phi(d_2)$
- Quoting units: vega per vol point, theta per calendar day; call and put with the same K , T share gamma and vega (parity)

Discrete Delta Hedging: the Algorithm

- Sell one call at the Black–Scholes price C_0 with volatility σ_{imp} ; buy Δ_0 shares; the rest is cash
- At each of N rebalancing dates: recompute Δ at the current price and time, buy or sell shares, update the cash with interest
- At expiry: **hedging error** = value of shares and cash minus the option pay-off
- Simulation: geometric Brownian motion, $\mu = 8\%$, $\sigma = 20\%$, three months, 20 000 paths
 - ▶ If the model is right and $\sigma_{imp} = \sigma$, the error vanishes as $N \rightarrow \infty$

Hedging Error and Rebalancing Frequency



□ Left: distribution of the hedging error for four frequencies; right: its standard deviation as % of the premium

 MFM_ch12_hedge_error

Reading the Hedging Error

- The mean error is close to 0 at every frequency: the premium is fair on average
 - ▶ Its standard deviation falls with N : monthly 40.8%, weekly 21.5%, daily 9.5%, four times a day 4.8% of the premium
- Log-log slope -0.48 : the error shrinks like $1/\sqrt{N}$; four times more trades halve it
- Worst 1% of paths: -2.68 with weekly and -1.17 with daily rebalancing, per option on a price of 100
- With transaction costs more trades cost more (a fixed fee per trade grows like N); the best frequency depends on the cost model and on how risk is traded off against cost

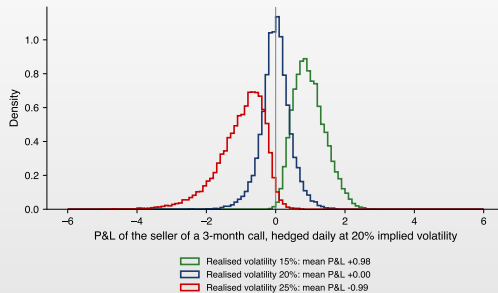
Why $1/\sqrt{N}$? The Discrete Hedging Error

- Over one interval $\Delta t = T/N$ the hedged book earns $\frac{1}{2}\Gamma S^2 [\sigma^2 \Delta t - (\Delta S/S)^2]$: time decay against realised squared return
- Mean 0 under the model; $\text{Var}[(\Delta S/S)^2] \approx 2\sigma^4 \Delta t^2$ and the terms are uncorrelated, so
 $\text{Var}(\text{error}) \approx \frac{1}{2}\sigma^4 \Delta t E \int_0^T \Gamma_t^2 S_t^4 dt$
- ATM call: $E[\Gamma_t^2 S_t^4] \approx \frac{K^2}{2\pi\sigma^2\sqrt{T^2 - t^2}}$ integrates to $K^2/(4\sigma^2)$, so s.d. $\approx \sigma K \sqrt{T/(8N)} = \sqrt{\pi/4} \mathcal{V}\sigma/\sqrt{N}$
 - With vega $\mathcal{V} = 19.72$ per unit: theory 1.01, 0.44, 0.22 vs simulated 0.96, 0.43, 0.22 for $N = 12, 63, 252$
- Asymptotic theory of the replication error: [Bertsimas, Kogan & Lo \(2000\)](#); it fails with jumps (the error does not vanish as $N \rightarrow \infty$), with stochastic volatility (vega risk) and with costs

Derivation: the Gamma-Weighted Volatility P&L

- Sell the option at σ_{imp} and hold $\Delta = \partial_S V^{imp}$; the price moves with σ_{real} : $dS = \mu S dt + \sigma_{real} S dW_t$
- Itô on V^{imp} : $dV = V_t dt + \Delta dS + \frac{1}{2} \Gamma \sigma_{real}^2 S^2 dt$; the equation at σ_{imp} gives $V_t = rV - rS\Delta - \frac{1}{2} \sigma_{imp}^2 S^2 \Gamma$
- Short option + Δ shares + cash at r , self-financing: $d \text{P\&L} = r \text{P\&L} dt + \frac{1}{2} \Gamma_t S_t^2 (\sigma_{imp}^2 - \sigma_{real}^2) dt$, so over the life
 - ▶ $\text{P\&L} = \int_0^T e^{r(T-t)} \frac{1}{2} \Gamma_t S_t^2 (\sigma_{imp}^2 - \sigma_{real,t}^2) dt$: path-dependent, weighted by Γ , large near K and near T
- A variance swap pays $\sigma_{imp}^2 - \sigma_{real}^2$ with constant weights; a delta-hedged option is a variance bet with random weights
- First-order size: vega $\times$ (implied $-$ realised); here vega = 0.197 per vol point, so 5 vol points $\approx$ 0.99

Selling Volatility at 20%: Three Outcomes

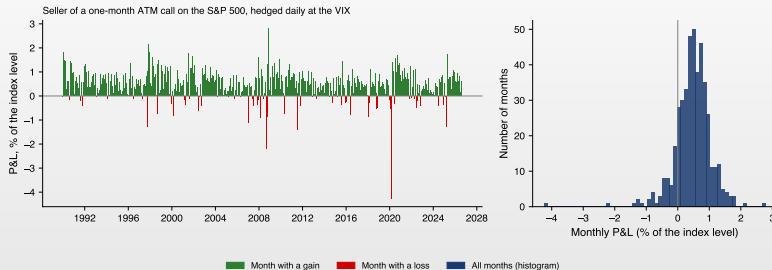


- Three-month call sold at 20% implied volatility, hedged daily; realised volatility 15%, 20% or 25%

Reading the Volatility Bet

- ▣ Realised 15%: mean P&L $+0.98$, positive on 99.8% of paths
- ▣ Realised 20%: mean $+0.00$: the premium was fair, only the discrete-hedging noise remains (s.d. 0.43)
- ▣ Realised 25%: mean -0.99 , positive on only 2.2% of paths
- ▣ Each result is close to $\text{vega} \times 5 \text{ vol points } (\pm 0.99)$: a delta-hedged option is a clean bet on volatility
- ▣ Next: which of the three worlds did the S&P 500 live in?

Thirty-Six Years of Selling S&P 500 Volatility



- Every 21 trading days: sell a one-month at-the-money call priced with the VIX, hedge it daily at the same volatility; P&L in % of the index; interest and dividends ignored

 MFM_ch12_delta_hedged_sp500

Reading the Delta-Hedged History

- ▣ 440 months from 1990-01: mean P&L 0.48% of the index per month, Newey–West $t = 16.0$ [Newey & West \(1987\)](#)
 - ▶ Positive in 86% of months; mean premium collected 2.22%
- ▣ The VIX exceeded the realised volatility of the month in 84% of months (means 19.3 vs 15.3)
- ▣ Worst month -4.25% (2020-03); skewness -1.46 : many small gains, a few large losses
- ▣ Stylised back-test: premiums come from the VIX, an option-strip index, not from quotes of the ATM call; the result is consistent with a **volatility risk premium**, but actual seller returns need historical option quotes and trading costs [Bakshi & Kapadia \(2003\)](#)

Recap: Pricing and Hedging

- Replication gives the Black–Scholes equation; the price depends on one unobserved number, σ
- Discrete hedging leaves an error $\approx \sqrt{\pi/4} \mathcal{V}\sigma/\sqrt{N}$; a wrong σ leaves the gamma-weighted variance P&L
- In a stylised VIX-priced back-test on the S&P 500, sellers of hedged options were paid on average, with crash risk: which volatility does the market use?

Implied Volatility

- **Implied volatility** σ_{imp} : the σ that makes the Black–Scholes price equal the market price
 - ▶ $BS(S, K, \tau, r, \sigma_{imp}) = C^{market}$; unique because vega > 0 : the price increases in σ
- Newton's method: $\sigma_{k+1} = \sigma_k - \frac{BS(\sigma_k) - C^{market}}{\mathcal{V}(\sigma_k)}$
 - ▶ Example: $S = K = 100$, three months, $r = 2\%$, market price 4.50; start at $\sigma_0 = 20.00\%$
 - ▶ $BS(\sigma_0) = 4.2322$, $\mathcal{V} = 19.848 \Rightarrow \sigma_1 = 21.3495\%$; the next step changes only the 6th decimal: $\sigma_{imp} = 21.3495\%$
- Traders quote options in implied volatility, not in money: it is comparable across strikes, maturities and assets
- If Black–Scholes were right, all options on one asset would have the **same** implied volatility

After October 1987: the Smile Appears



Amsterdam stock exchange after the crash, 21 October 1987

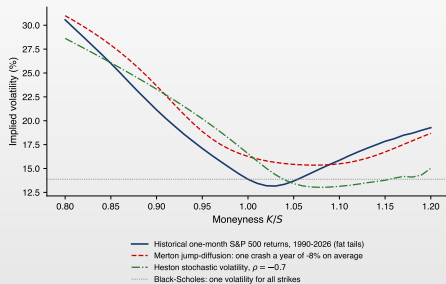
Photo: [Bart Molendijk / Anefo \(1987\)](#); CC0; [Wikimedia Commons](#)

- Before 1987 index options had a nearly flat implied volatility across strikes [Rubinstein \(1994\)](#)
 - ▶ Since the crash, low-strike puts are much more expensive: the skew
- Option prices showed crash fears from October 1986 to August 1987, but not strongly in the two months immediately before the crash [Bates \(1991\)](#)
- **Smile**: implied volatility as a function of K ; **term structure**: as a function of T ; **surface**: both
- Crash risk on the Bucharest Stock Exchange: [Pele & Mazurencu-Marinescu \(2012\)](#)

Why Do Smiles Exist?

- **Fat tails:** large moves are more frequent than the Normal distribution allows (Chapter 1)
 - ▶ Deep OTM options pay only after large moves, so they are worth more than Black–Scholes says
- **Leverage and volatility feedback:** when prices fall, volatility rises [Christie \(1982\)](#)
 - ▶ Negative skewness of returns: the left tail is longer, so low-strike puts are dearer
- **Crash fear and demand for insurance:** investors pay a premium for puts [Bates \(1991\)](#)
 - ▶ Intermediaries who sell them need compensation for crash risk they cannot hedge
- Models: jumps [Merton \(1976\)](#), stochastic volatility [Heston \(1993\)](#), [Hull & White \(1987\)](#), local volatility [Gatheral \(2006\)](#)

Three Sources of a Smile

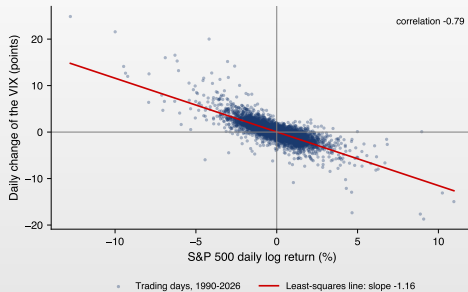


- One-month implied volatility from: the historical distribution of S&P 500 monthly returns (rescaled to mean S , a martingale with $r = 0$, no risk adjustment), a jump model and a stochastic-volatility model

Reading the Model Smiles

- ▣ Historical S&P 500: 9 225 overlapping one-month returns, skewness -1.27 , excess kurtosis 6.8
 - ▶ Implied volatility 30.6% at $K/S = 0.8$, 13.9% at the money, 19.3% at 1.2 : fat tails and negative skewness together create a skewed smile
- ▣ Merton jumps (one crash of about -8% a year): 31.0% vs 16.3% : jumps steepen short-dated smiles
- ▣ Heston with $\rho = -0.7$: 28.6% vs 16.6% vs 13.1% : negative correlation between price and variance tilts the smile into a skew
- ▣ This benchmark has no risk adjustment; risk-neutral index distributions are more left-skewed than realised ones [Jackwerth \(2000\)](#), so priced crash risk steepens market smiles further

The Leverage Effect in the Data



- Daily S&P 500 log return vs the change of the VIX on the same day, 1990–2026

Reading the Leverage Effect

- ▣ Correlation -0.79 over 9 244 days: implied volatility rises when the market falls
- ▣ Slope -1.16 VIX points per 1% return; steeper on down days (-1.39) than on up days (-1.01)
 - ▶ Mean VIX change: $+0.88$ on down days, -0.76 on up days
- ▣ This is the $\rho < 0$ of the Heston model and the asymmetry of GJR-GARCH (Chapter 5)
- ▣ A long position in low-strike puts gains from both the fall and the rise in volatility: this double hedge is why they are expensive

Models of the Smile

Model	Idea	Fits best
Jumps Merton (1976)	Poisson jumps added to the Brownian motion	steep short-dated smiles
Stochastic volatility Heston (1993)	variance follows its own mean-reverting process, correlated with S	skew and term structure of longer maturities
Local volatility Gatheral (2006)	$\sigma(S, t)$ read off the whole surface (Dupire's formula)	matches every quoted price today, exotic options
Parametric smile, SVI Gatheral & Jacquier (2014)	five parameters per expiry, no dynamics	interpolation and quoting (next section)

- Chapter 11 introduces the jump and stochastic-volatility processes; [Franke, Härdle & Hafner \(2019\)](#) and [Čížek, Härdle & Weron \(2011\)](#) give the implied-volatility surface in detail

Recap: Implied Volatility

- ▣ Implied volatility is the market price of an option translated into one number
- ▣ It varies with the strike (smile, skew) and the maturity (term structure): Black–Scholes is a quoting convention, not a model of the data
- ▣ Fat tails, leverage and crash fear all tilt the smile towards low strikes
- ▣ Next: a full surface from a real market, Bitcoin options

Bitcoin Options on Deribit

- Deribit lists European options on Bitcoin and Ether, traded 24/7
 - ▶ **Inverse** options: premium and pay-off are paid in Bitcoin; the price in USD is the BTC premium times the Bitcoin price
- Daily, weekly, monthly and quarterly expiries; each expiry has its own forward price F
- DVOL: Deribit's 30-day implied-volatility index for Bitcoin [Deribit DVOL](#)
- Our snapshot: 22 September 2026, 07:51 UTC; 9 expiries with at least two days left, 375 quoted OTM options
 - ▶ Cryptos as an asset class and their risk: [Pele et al. \(2023\)](#); hedging Bitcoin options: [Matic, Packham & Härdle \(2023\)](#)

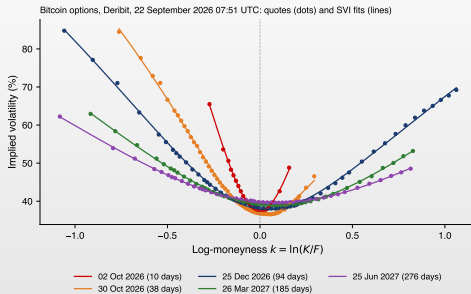
A Parametric Smile: SVI

- Work with the **total implied variance** $w(k) = \sigma_{imp}^2(k) \tau$ at log-moneyness $k = \ln(K/F)$
 - ▶ SVI (stochastic volatility inspired) [Gatheral & Jacquier \(2014\)](#): $w(k) = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + s^2} \right)$
- Parameters: a level, b slope of the wings, ρ tilt (negative = put skew), m shift, s curvature at the bottom
 - ▶ Wing slopes $b(1 \pm \rho)$: [Lee \(2004\)](#) bounds the asymptotic slope, $\limsup_{|k| \rightarrow \infty} w(k)/|k| \leq 2$, so $b(1 + |\rho|) \leq 2$ is necessary; $w \geq 0$ needs $a + bs\sqrt{1 - \rho^2} \geq 0$
- No arbitrage: call prices must fall and be convex in K (no butterfly arbitrage) and w must increase with τ (no calendar arbitrage)
 - ▶ One SVI fit per expiry by least squares; the exact conditions are on the next slide

No-Arbitrage Constraints on the Smile

- **Wings**, moment formula of [Lee \(2004\)](#): $\beta_R = \limsup_{k \rightarrow \infty} w(k)/k \in [0, 2]$ and $\tilde{p} = \frac{1}{2\beta_R} + \frac{\beta_R}{8} - \frac{1}{2}$
 - ▶ $\tilde{p} = \sup\{p : E^Q[S_T^{1+p}] < \infty\}$: a steeper wing means fatter tails; the left wing mirrors it with negative moments
- **Butterfly** (density $q \geq 0$), with $w' = \partial_k w$ [Gatheral & Jacquier \(2014\)](#):
 - ▶ $g(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{w'^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \geq 0$ for all k
 - ▶ and $d_+(k) = -k/\sqrt{w} + \sqrt{w}/2 \rightarrow -\infty$ as $k \rightarrow \infty$ ([Gatheral & Jacquier \(2014\)](#), Lemma 2.2); for SVI this follows from a right-wing slope below 2
- **Calendar**: $\partial_\tau w(k, \tau) \geq 0$ for every k : total variance cannot fall with maturity
- Slice-by-slice least squares imposes none of them; SSVI, the surface version of SVI, has simple sufficient conditions ([Gatheral & Jacquier \(2014\)](#), Theorems 4.1–4.2)

The Bitcoin Smile



- Mark implied volatilities of OTM options (dots) and SVI fits (lines) for five expiries

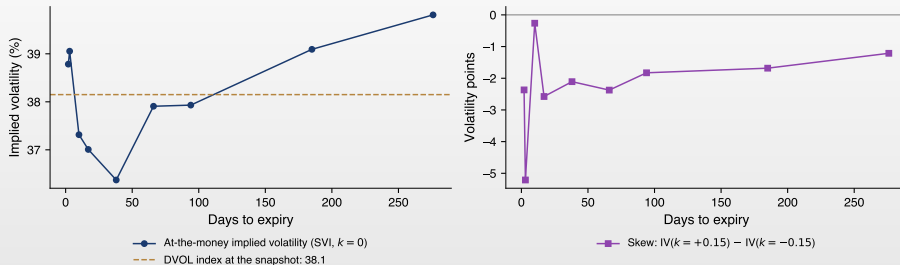
Reading the Bitcoin Smile

- ▣ A true **smile**: both wings rise, unlike the one-sided equity skew
 - ▶ Bitcoin can jump up as well as down: call buyers pay for rallies, put buyers for crashes [Zulfiqar & Gulzar \(2021\)](#)
- ▣ Short expiries have the steepest wings: a jump over a few days matters more relative to $\sigma\sqrt{\tau}$
- ▣ SVI fits the quotes closely: root mean squared error at most 0.76 vol points per expiry
- ▣ At this snapshot puts are slightly dearer than calls: skew from -5.2 to -0.3 vol points

Checking the Nine Bitcoin Fits

- ▣ Wings: largest slopes 0.31 (right) and 0.39 (left), far below 2: Lee's bound holds
 - ▶ Read through Lee: right slope 0.31 $\Rightarrow \tilde{\rho} = 1.18$; a property of SVI's linear extrapolation beyond the last strike, not of the data
- ▣ Butterfly: $\min_k g(k) > 0$ on the grid $k \in [-1.5, 1.5]$ for all 9 expiries (lowest 0.036, 27 November 2026); slopes below 2 give the tail condition
- ▣ Calendar: 1 of 8 adjacent pairs crosses: $w(2 \text{ October } 2026) > w(9 \text{ October } 2026)$ on 24% of $k \in [-0.5, 0.5]$, by up to -0.0075
 - ▶ A calendar arbitrage created by the estimator, not quoted by the market: separate fits ignore the cross-expiry restriction
- ▣ Remedy: estimate the surface jointly under the constraints (SSVI, or least squares with $g \geq 0$ and $\partial_\tau w \geq 0$)

Term Structure and Skew



- Left: ATM implied volatility by expiry and the DVOL index; right: skew = implied volatility at $k = +0.15$ minus at $k = -0.15$

Reading the Term Structure

- ▣ ATM implied volatility between 36.4% and 39.8%; the longest expiry (276 days): 39.8%
 - ▶ DVOL 38.1 vs ATM 37.0% and 36.4% at the expiries around 30 days; a variance-strip index need not lie between ATM volatilities
- ▣ An upward-sloping term structure: the market expects calm now and more uncertainty later
- ▣ The skew is negative for 9 of 9 expiries and closest to 0 at long maturities
- ▣ A snapshot is one day: the level and the sign of the skew change with the market state

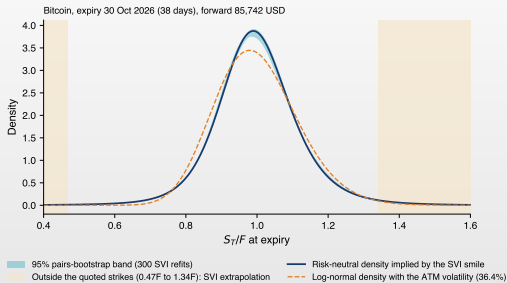
The Risk-Neutral Density

- ▣ Breeden & Litzenberger (1978): the second derivative of call prices in the strike gives the risk-neutral density q of S_T
 - ▶ $q(K) = e^{r\tau} \frac{\partial^2 C(K)}{\partial K^2}$; a butterfly spread $C(K-h) - 2C(K) + C(K+h)$ pays like a narrow bet on $S_T \approx K$
- ▣ Needs a smooth smile: SVI gives $C(K)$ at every strike, then differentiate twice
- ▣ Convexity of C in K (no butterfly arbitrage) is exactly the condition $q \geq 0$
- ▣ q mixes probabilities and risk aversion: comparing it with the physical density gives the **pricing kernel** Aït-Sahalia & Lo (1998), Grith, Härdle & Park (2013)

Estimating the Density: a Statistical Problem

- $q = e^{r\tau} \partial^2 C / \partial K^2$ from noisy, discrete quotes is an ill-posed inverse problem: differentiating twice amplifies the noise
- Estimators: fit a smile (SVI) and differentiate; kernel regression of C on (K, τ) [Aït-Sahalia & Lo \(1998\)](#); the same under monotonicity and convexity in K [Aït-Sahalia & Duarte \(2003\)](#)
 - ▶ Smoothness-penalised implied trees [Jackwerth & Rubinstein \(1996\)](#); Härdle's shape-invariant pricing kernels [Grith, Härdle & Park \(2013\)](#)
- Uncertainty: pairs bootstrap of the quotes, refit, differentiate: pointwise bands for q and intervals for tail probabilities
- Beyond the last quoted strike the data identify nothing: the tail is an extrapolation (SVI wings or an extreme-value tail); report where the quotes end
- Pricing kernel $\propto q/p$ with p the physical density: its non-monotone shape is the pricing-kernel puzzle [Jackwerth \(2000\)](#)

What the Market Prices for Bitcoin



- Risk-neutral density from the SVI smile of the expiry closest to 30 days with a 95% pairs-bootstrap band, vs the log-normal density with the ATM volatility; shaded: outside the quoted strikes

Reading the Risk-Neutral Density

- ▣ Expiry in 38 days, forward 85 742 USD, ATM volatility 36.4%; 56 quotes from $0.47F$ to $1.34F$
 - ▶ Taller centre and fatter tails than the log-normal density: the smile in density form
- ▣ Fall of more than 30%: 1.21%, 95% bootstrap interval [1.08, 1.24]%, vs 0.14% log-normal: about 8 times more
- ▣ Fall of more than 20%: 4.28% [4.04, 4.46] vs 3.27%; rise above 20%: 5.3% vs 5.3%, almost log-normal
- ▣ The band is narrow ($\pm 2.5\%$ at the centre, $\pm 8\%$ at $0.7F$) because the quotes are the exchange's smoothed mark volatilities
 - ▶ It measures fitting noise, not bid-ask uncertainty; $0.7F$ lies inside the quoted strikes, but the probability stays conditional on SVI interpolation and extrapolation: 0.13% of the 1.21% lies below the last quote
- ▣ Prices of insurance, not forecasts: they include the premium for bearing crash risk

Recap: the Volatility Surface

- ▣ A surface is an estimate: one SVI per expiry, checked against Lee's bound, $g(k) \geq 0$ and $\partial_\tau w \geq 0$
- ▣ The risk-neutral density is a second derivative of prices: report bands and where the quotes end
- ▣ Bitcoin: both wings rich, crash probability many times the log-normal one

Variance Swaps

- A **variance swap** pays at T : notional $\times$ (realised variance – strike K_{var}), with realised variance $\frac{252}{n} \sum r_t^2$
 - ▶ The fair strike is the risk-neutral expected variance; with continuous price paths options span it without a model for σ_t (next slides) [Demeterfi, Derman, Kamal & Zou \(1999\)](#)
- Spanning with $f = -2 \ln(S/F)$ (Section 2): a strip of OTM options weighted by $1/K^2$ plus a futures position
 - ▶
$$K_{var} = \frac{2e^{r\tau}}{\tau} \left(\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right)$$
- Low strikes get the largest weights: expensive puts make variance expensive
- Buyer of variance = insurance against turbulence; seller = earns a premium, pays in crashes [Carr & Wu \(2009\)](#)

Model-Free Implied Variance: What the Strip Measures

- Itô on $\ln S$ (continuous paths): $-2 \ln(S_T/F) = \int_0^T \sigma_t^2 dt - 2 \int_0^T dS_t/S_t$ (with F the forward, $r = q = 0$ for simplicity)
 - ▶ The last term is a Q -martingale, so the strip prices $E^Q[\int_0^T \sigma_t^2 dt]$ for any diffusion [Britten-Jones & Neuberger \(2000\)](#)
- With jumps J : the strip prices $\int \sigma_t^2 dt + 2 \sum (e^J - 1 - J)$, but realised variance is $\int \sigma_t^2 dt + \sum J^2$
 - ▶ Gap $2E^Q \sum (e^J - 1 - J - J^2/2) \approx E^Q \sum J^3/3$: negative for crashes, so the strip understates expected realised variance [Carr & Wu \(2009\)](#)
- Merton model of Section 6 (one jump a year, mean -8%): strip 17.82 vs $\sqrt{E^Q[QV]} = 18.00$ vol points, -2.0% in variance
- “Model-free” means free of a model for σ_t , not of continuity
- SVIX [Martin \(2017\)](#), from $f = S^2$: $SVIX_t^2 = \frac{2}{\tau R_f S_t^2} \left[\int_0^F P dK + \int_F^\infty C dK \right] = \text{var}_t^Q(R_T/R_f)/\tau$, with R_T, R_f gross returns
 - ▶ If $\text{cov}_t(M_T R_T, R_T) \leq 0$ (M_T the stochastic discount factor): $\frac{1}{\tau}(E_t R_T - R_f) \geq R_f SVIX_t^2$, a lower bound on the equity premium

The VIX Index

- Cboe Volatility Index (VIX), since 1993; since 2003 computed from S&P 500 options with the variance-swap formula [Cboe \(2025\)](#)
 - ▶ $\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2$, K_0 = first strike at or below F ; Q = bid-ask mid of the put for $K_i < K_0$, of the call for $K_i > K_0$, the average of both at K_0
 - ▶ $\Delta K_i = (K_{i+1} - K_{i-1})/2$, one-sided at the two ends; options with a zero bid are dropped, and each wing stops after two consecutive zero bids
- Two expiries around 30 days, interpolated to exactly 30 days; $VIX = 100 \sigma$
- VIX futures since 2004, VIX options since 2006 [Cboe \(2025\)](#); the “fear gauge” [Whaley \(2000\)](#)
- Siblings: VIX9D (9 days), VIX3M (3 months), VVIX (the VIX of the VIX)

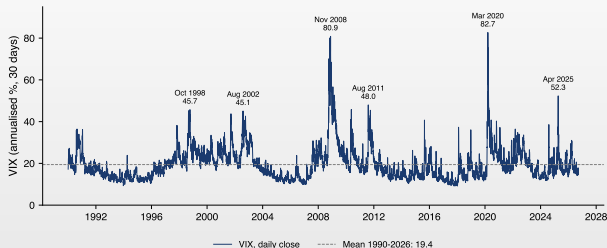
Worked Example: a VIX for Bitcoin

- Cboe rules on our Deribit snapshot: USD mid $Q = \text{BTC mid} \times \text{index } S; F$ and $e^{RT} = F/S$ from Deribit's forward (Cboe: F from parity, R from Treasury yields)
 - ▶ Near expiry: 17 days, 26 OTM options, $\sigma = 39.5\%$
 - ▶ Next expiry: 38 days, 56 OTM options, $\sigma = 40.2\%$
- Interpolate the total variances $\sigma_i^2 T_i$ linearly in time to 30 days:
 - ▶ $\sigma_{30}^2 = \frac{1}{30} \left[T_1 \sigma_1^2 \frac{T_2 - 30}{T_2 - T_1} + T_2 \sigma_2^2 \frac{30 - T_1}{T_2 - T_1} \right]$ (days)
- Our Bitcoin VIX: 40.0; Deribit's DVOL at the same moment: 38.1 [Alexander & Imeraj \(2020\)](#)
- Both exceed the ATM volatility of the two expiries: the variance swap also pays for the wings of the smile

Truncation and Discretisation: How Much Do They Matter?

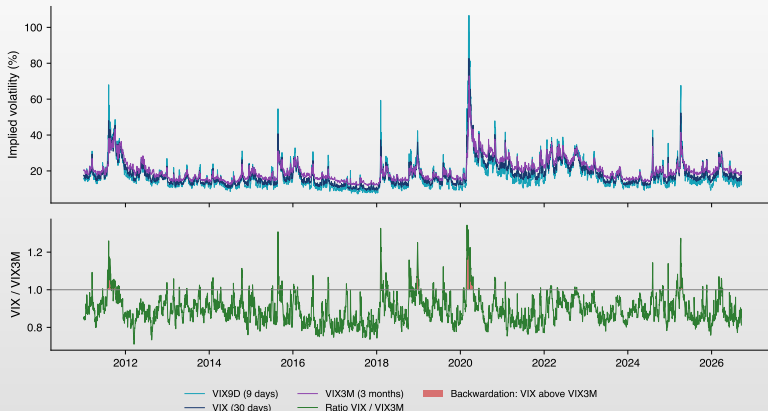
- Only a finite set of strikes is quoted: truncating the strip biases the index down, a coarse grid biases it up [Jiang & Tian \(2005\)](#)
- Bitcoin, same two expiries, 30-day index: quoted strikes 40.03; SVI prices at the same strikes 39.99
 - ▶ SVI on a dense grid inside the quoted range 39.78; extended to $k \in [-4, 4]$ with the SVI wings 40.34
- The near expiry is quoted only from $0.76F$ to $1.17F$: its volatility rises from 39.05 to 40.00 when the wings are added
- These choices move the index by at most 0.6 vol points, less than the gap to DVOL (1.9): the rest reflects the quote selection and methodology of DVOL, not quantified here; an index is an estimator, so report its sensitivity
- A log-normal benchmark with known answer: Seminar 12, A8

The VIX, 1990–2026



- Daily close with the six highest peaks at least 500 days apart
- Mean 19.4 (5.6% a month), range 9.1–82.7 (2020-03-16, COVID-19); above 30 on 7.9% of days: sudden spikes, slow decay, as in GARCH (Chapter 5)

The VIX Term Structure

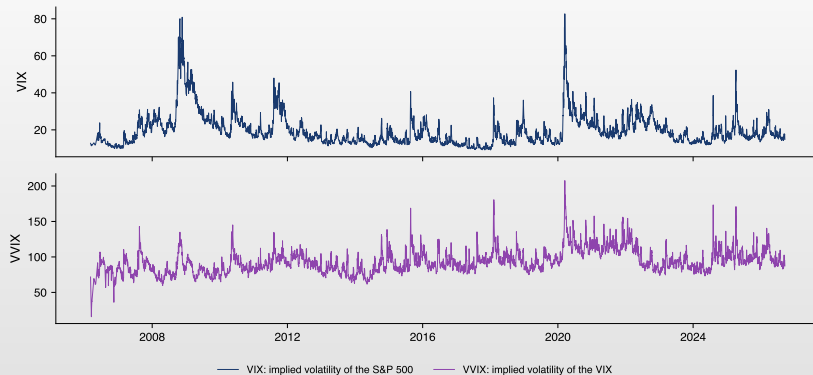


Top: implied volatility over 9 days, 30 days and 3 months; bottom: ratio VIX / VIX3M, 2011–2026

Reading the Term Structure

- Normally upward sloping (**contango**): $\text{mean VIX} / \text{VIX3M} = 0.89$; VIX9D below the VIX on 73% of days
 - ▶ In calm markets short-term uncertainty is low and investors pay more for longer protection
- Inverted (**backwardation**) on 7.6% of days, always in crises: fear is concentrated in the near term
- Next 21 days after backwardation: S&P 500 return 2.88% and realised volatility 26.7%, vs 0.80% and 13.6% in contango
- High risk, high expected return after panics: the premium for bearing crash risk [Bollerslev, Tauchen & Zhou \(2009\)](#)

The Volatility of Volatility



□ VIX (top) and VVIX (bottom), daily closes, 2006–2026

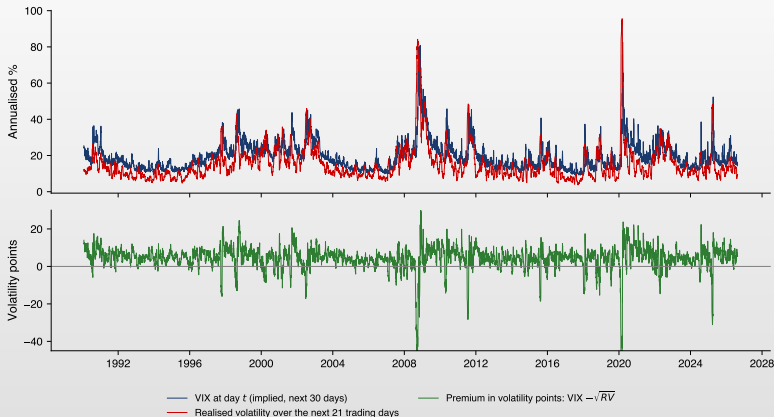
Reading the VVIX

- Mean VVIX 93.5: the VIX itself is expected to move about 5.3 times more than the index, in relative terms
- Highest value 207.6 on 2020-03-16, the same day as the VIX peak
- Daily changes of VIX and VVIX correlate 0.71: when fear rises, uncertainty about fear rises too
- Volatility is itself random: the core of stochastic-volatility models such as Heston [Heston \(1993\)](#)

The Variance Risk Premium

- Variance risk premium (VRP) Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009); two versions that must not be confused:
 - ▶ **Economic (ex ante):** $E_t^Q[RV_{t,t+21}] - E_t^P[RV_{t,t+21}]$; VIX_t^2 estimates the first term, the second needs a forecast
 - ▶ **Ex post** (used here): $VIX_t^2 - RV_{t,t+21}$, the realised P&L of a variance swap, known only at $t + 21$
 - ▶ **Ex-ante proxy:** $VIX_t^2 - RV_{t-21,t}$, past variance as the forecast of E_t^P ; known at t , so it can serve as a predictor (last slides of this section); other forecasts give other proxies
 - ▶ $RV_{t,t+21} = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2$ (annualised, in %²)
- Positive on average if investors dislike states with high volatility, which are also bad states for wealth
- Overlapping 21-day windows are autocorrelated: Newey–West with at least 21 lags Newey & West (1987); persistent volatility can extend the dependence, so check longer bandwidths

Implied vs Realised Volatility, S&P 500

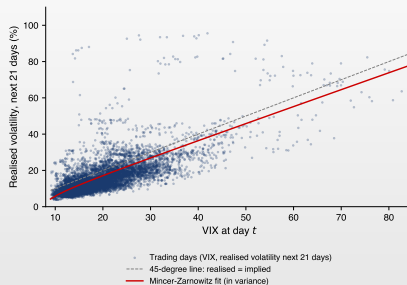


Top: VIX and the volatility realised over the next 21 trading days; bottom: their difference, 1990–2026

Reading the Variance Risk Premium

- ▣ 9 203 days: mean VIX 19.4, mean realised volatility 15.3
 - ▶ Premium 4.09 vol points (Newey–West s.e. 0.27); in variance 113 %², $t = 5.5$
- ▣ Positive on 86% of days: implied volatility is usually too high as a forecast
- ▣ The exceptions are the crashes: worst -72.0 vol points, for the window starting 2020-02-20 (COVID-19)
- ▣ Same message as the delta-hedged option: sellers of volatility are paid on average and lose in crises

Is the VIX a Good Forecast?



- Realised volatility over the next 21 days vs the VIX, with the 45-degree line and the Mincer–Zarnowitz fit [Mincer & Zarnowitz \(1969\)](#)

Reading the Forecast Regression

- $RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon_t$ Mincer & Zarnowitz (1969): an unbiased forecast has $\alpha = 0$, $\beta = 1$; Newey–West errors, 21 lags
 - ▶ Intercept $\hat{\alpha} = -51.9$ (34.4), $\hat{\beta} = 0.86$ (0.10), $R^2 = 0.37$
- One at a time neither differs significantly from (0, 1): $t = -1.51$ and -1.34
 - ▶ But the estimates are correlated -0.90 : the joint Wald test gives $\chi^2_2 = 41.7$, $p = 8.9 \cdot 10^{-10}$: unbiasedness is rejected jointly
- Most points lie below the 45-degree line: the VIX overstates future volatility by the risk premium
- Past realised variance alone: $R^2 = 0.29$; with both, the VIX keeps its weight (0.77) and past variance adds little (0.08, s.e. 0.13)

Is the VIX an Efficient Forecast? Inference Done Right

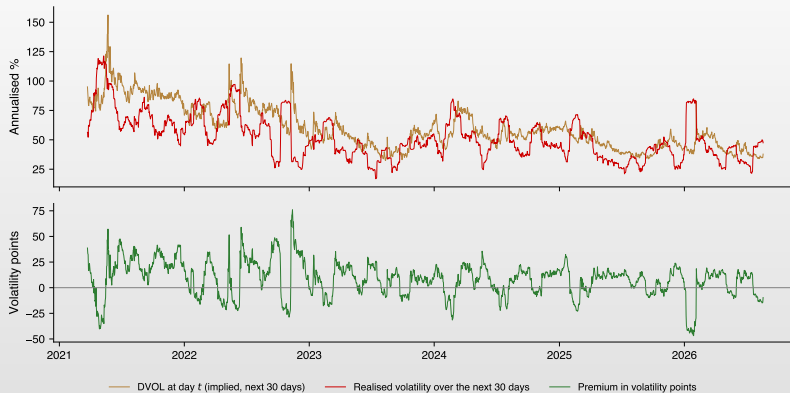
Specification	$\hat{\beta}$	s.e.	$t(\beta = 1)$	Note
Levels, OLS	0.86	0.10	-1.34	joint Wald $p = 8.9 \cdot 10^{-10}$
Logs, OLS	1.06	0.035	1.65	intercept -0.94, $R^2 = 0.57$
Logs, instrumental variables	1.02	0.042	0.60	first-stage $t = 37.7$

- Levels are dominated by 2008 and 2020; in logs the slope is close to 1 and the premium is a proportional shift (intercept < 0)
- VIX^2 measures expected variance with error, which attenuates OLS: instrument it with VIX^2 21 days earlier [Christensen & Prabhala \(1998\)](#); here the slope barely moves
 - ▶ Valid only if lagged VIX^2 is uncorrelated with the forecast error (exclusion, untestable with one instrument): a sensitivity check, not proof
- Newey–West errors with 21 lags; instrumental variables estimated by just-identified GMM with a HAC covariance

Does the VIX Beat HAR-RV? Encompassing and Loss-Based Tests

- Heterogeneous autoregressive model (HAR-RV) [Corsi \(2009\)](#): past daily, weekly and monthly realised variance; encompassing regression in logs [Busch, Christensen & Nielsen \(2011\)](#)
 - ▶ VIX slope 0.90 (0.056); HAR terms jointly $\chi^2_3 = 14.2$, $p = 0.003$; R^2 : VIX 0.570, HAR 0.474, both 0.575
- Out of sample from 2000 (6 697 days; coefficients re-estimated every 21 days on targets already observed)
 - ▶ QLIKE loss $L = RV/\hat{\sigma}^2 - \ln(RV/\hat{\sigma}^2) - 1$, robust to noise in the RV proxy [Patton \(2011\)](#)
 - ▶ Loss ratio VIX^2 / HAR : 0.94, Diebold–Mariano $t = -0.93$ ($p = 0.35$) [Diebold & Mariano \(1995\)](#); MZ-corrected VIX: 0.95 ($p = 0.35$)
- In sample, history adds significant information; out of sample the VIX is about 6% better but not significantly: the evidence is weaker than the in-sample tests suggest
- Targets overlap by 20 of 21 days: the Diebold–Mariano variance is HAC with 21 lags

The Variance Risk Premium of Bitcoin



- DVOL and the Bitcoin volatility realised over the next 30 calendar days, 2021–2026; bottom: their difference

Reading the Bitcoin Premium

- 1975 days: mean DVOL 60.3 vs realised 51.7; premium 8.5 vol points (s.e. 1.6, $t = 5.3$)
 - ▶ Twice the S&P 500 premium in vol points, but similar relative to the level of volatility
- Positive on 71% of days: less often than for the S&P 500; long episodes of negative premium
- Worst: -46.9 vol points for the window starting 2026-01-28
- A young market and a short sample: 66 non-overlapping 30-day periods; a Bitcoin VIX and its premium:
Alexander & Imeraj (2020)

Does the Ex-Ante VRP Predict Returns? Overlapping-Horizon Inference

Horizon	$\hat{b}$	t_{NW}	t_{HH}	$t_{Hodrick}$	p_{boot}	R^2 (%)
$h = 1$	0.07	0.33	0.36	0.41	0.35	0.2
$h = 3$	0.09	0.69	0.67	1.33	0.07	1.2
$h = 6$	0.03	0.49	0.48	0.65	0.26	0.3
$h = 12$	-0.02	-0.64	-0.79	-0.54	0.69	0.2

- Monthly 1990–2026: $\frac{12}{h} \sum_{j=1}^h r_{t+j}^e = a + b \text{VRP}_t + u_t$, $\text{VRP}_t = \text{VIX}_t^2/12 - \text{RV}_t$, ex-ante proxy with past RV_t as the forecast [Bollerslev, Tauchen & Zhou \(2009\)](#)
- If one-month errors are serially uncorrelated, overlap makes u_t an $\text{MA}(h-1)$: Newey–West (h lags), Hansen–Hodrick [Hansen & Hodrick \(1980\)](#), Hodrick 1B [Hodrick \(1992\)](#) (better size); these bandwidths are choices, so check their sensitivity
- p_{boot} : one-sided, VRP simulated as $\text{AR}(1)$ ($\hat{\phi} = 0.13$) under $b = 0$, residuals resampled jointly [Stambaugh \(1999\)](#); low persistence, so little small-sample bias
- No horizon is significant on the full sample; the 1990–2007 evidence is revisited in the last section

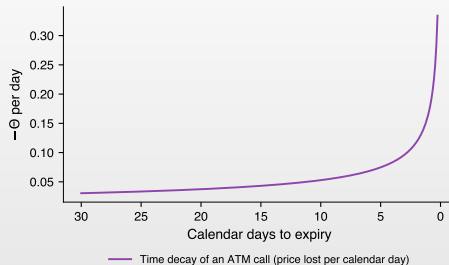
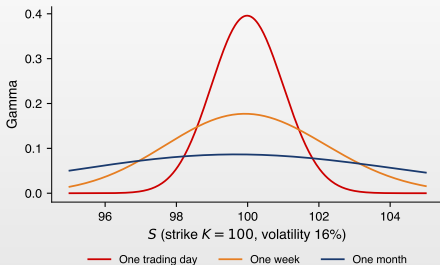
Recap: Implied Volatility as Information

- The VIX is an annualised volatility index: 100 times the square root of an option-strip estimate of the fair strike of a 30-day variance swap on the S&P 500
- Its term structure flags crises (backwardation); VVIX measures uncertainty about volatility
- Implied variance exceeds realised variance on average, for equities and for Bitcoin: the variance risk premium
- As a forecast, the VIX is biased upwards in levels, has slope near 1 in logs, and beats HAR-RV only insignificantly out of sample

Options That Live One Day

- ▣ **0DTE** (zero days to expiry) options: traded on the day they expire [Cboe 0DTE](#)
 - ▶ S&P 500 options now expire every trading day, so a 0DTE contract exists every day
- ▣ More than 75% of retail trades in S&P 500 options involve 0DTE contracts, and retail investors lose money on them [Beckmeyer, Branger & Gayda \(2023\)](#)
 - ▶ 0DTE options were 48% of index-option trading in August 2023 [Brogaard, Han & Won \(2023\)](#)
- ▣ More 0DTE trading is followed by higher index volatility, mostly through speculative retail trades [Brogaard, Han & Won \(2023\)](#); ultra-short pricing needs skewness and kurtosis corrections [Bandi, Fusari & Renò \(2023\)](#)
- ▣ Theory from Section 4: with $\tau \rightarrow 0$ gamma and theta explode at the money

Gamma and Time Decay Near Expiry



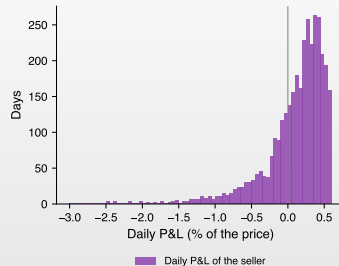
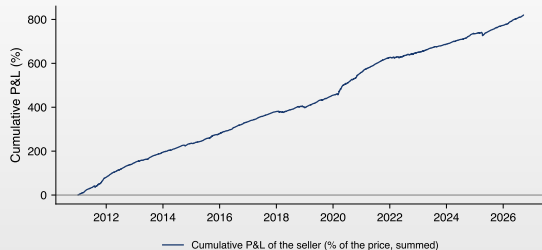
□ Left: gamma of a call with $K = 100$ one day, one week and one month before expiry; right: daily time decay of an ATM call

 MFM_ch12_0dte_gamma

Reading Gamma Near Expiry

- ▣ ATM gamma one day before expiry: 0.396 vs 0.086 one month before: 4.6 times larger ($= \sqrt{21}$)
- ▣ A 1% move in the last day changes delta by about 0.4: the hedge must be adjusted many times within the day
- ▣ Time decay per day: 0.167 in the last day vs 0.030 with 30 days left, about 5.5 times more
- ▣ The seller collects fast time decay and carries large gamma risk: small premium, sudden losses

Selling a Same-Day Straddle: a Stylised Test



- SPY, 2011–2026: sell an ATM straddle at the open, settle at the close; premium = $\sqrt{2/\pi} \sigma_{day} S$ with σ_{day} from the previous VIX9D scaled to the trading session

 MFM_ch12_0dte_straddle

Reading the Straddle Test

- ▣ Session share of daily variance 0.74 (estimated on 1993–2010); 3 950 days
 - ▶ Mean premium 0.76% of the price vs mean absolute open-to-close move 0.55%
- ▣ The seller wins on 75% of days, 20.7 basis points a day on average
- ▣ Worst day -8.25% (2025-04-09), about 11 premia; the move exceeds twice the premium on 3.5% of days
- ▣ Stylised pricing, not traded prices: the result measures the variance premium inside VIX9D
 - ▶ Market 0DTE premia, spreads and intraday hedging would change the numbers [Bandi, Fusari & Renò \(2023\)](#), [Beckmeyer, Branger & Gayda \(2023\)](#)

Case Study: the Research Question

- **Question Back, Crotty & Kazempour (2022):** option prices give lower bounds on the equity premium; are they valid and tight conditionally, and do they forecast returns out of sample?
 - ▶ Why: a tight bound is a real-time equity-premium forecast with no estimated parameter; **Martin (2017)** could not reject $\alpha = 0$, $\beta = 1$ on 1996–2012, a test with low power
- **Martin bound** (eq. 3), $R_{t,T}$ the gross market return, $R_{f,t,T}$ the gross riskless return, var_t^* the risk-neutral variance:
 - ▶ $E_t[R_{t,T}] - R_{f,t,T} \geq R_{f,t,T} \text{SVIX}_{t,T}^2$, $\text{SVIX}_{t,T}^2 = \text{var}_t^*(R_{t,T}/R_{f,t,T})$
 - ▶ Valid if $\text{cov}_t(M_{t,T}R_{t,T}, R_{t,T}) \leq 0$ ($M_{t,T}$ the stochastic discount factor); tight if this covariance is zero, e.g. a log-utility investor who holds the market (Section 1)
- **Chabi-Yo–Loudis bound** **Chabi-Yo & Loudis (2020)**: built from the first four risk-neutral moments, free of preference parameters in its restricted form; almost always above the Martin bound (Figure 1)
- **Link to this chapter:** SVIX is the static-spanning strip with $f = S^2$ (Sections 2 and 8)
 - ▶ Realised slackness $s_{t,T} = R_{t,T}^e - b_{t,T}$ (excess return minus bound): valid means $E_t[s_{t,T}] \geq 0$, tight means $E_t[s_{t,T}] = 0$

Case Study: Data and Method

- **Data** (Section 3, Appendix C): daily Martin and Chabi-Yo–Loudis bounds, January 1990 – December 2020, from S&P 500 index options (Cboe data for 1990–1995 and 2020, OptionMetrics for 1996–2019)
 - ▶ Filters: standard options only (no weekly or quarterly), bid > 0, the OTM option at each strike, 7 to 550 days, at least 10 strikes per expiry; bounds interpolated to 30, 90, 180 and 360 days
 - ▶ Returns: S&P 500 with dividends minus the Fama–French risk-free rate, over 21, 63, 126 and 252 trading days up to March 2021, annualised, in %
- **Moments** (eq. 10): for any nonnegative z_t known at t , $E[(R_{t,T}^e - b_{t,T}) z_t] \geq 0$; z_t = a constant and nine Welch–Goyal predictors made positive [Welch & Goyal \(2008\)](#) (Table 3)
- **Kodde–Palm tests** [Kodde & Palm \(1986\)](#) (eqs. 11–14): $\bar{\lambda}$ the 10 sample means, Σ their Hansen–Hodrick covariance [Hansen & Hodrick \(1980\)](#), with lags equal to the days in the horizon
 - ▶ $D_1 = \min_{\lambda \geq 0} (\lambda - \bar{\lambda})^\top \Sigma^{-1} (\lambda - \bar{\lambda})$ tests validity ($H_0: \lambda_0 \geq 0$); $D_2 = \bar{\lambda}^\top \Sigma^{-1} \bar{\lambda} - D_1$ tests tightness ($H_0: \lambda_0 = 0$ vs $\lambda_0 \geq 0$); both follow mixtures of χ^2 laws
- **Forecasting**: bias-corrected predictive regressions [Amihud & Hurvich \(2004\)](#) (Tables 5, 6); monthly out-of-sample forecasts against the expanding post-1926 market mean, 60-month initial window (Table 8)
 - ▶ $R_{OS}^2 = 1 - \sum \varepsilon_t^2 / \sum \nu_t^2$, ε_t and ν_t the errors of the forecast and of the benchmark; Diebold–Mariano tests [Diebold & Mariano \(1995\)](#)

Case Study: Key Results (Tables 1, 3, 5, 6 and 8 of the Paper)

Horizon	Mean R^e (%)	Mean b (%)	D_1	D_2	$\hat{\beta}$ (s.e.)	R_{OS}^2 (%)
1 month	8.76	3.81	0.00	29.93	1.36 (1.23)	−1.2
3 months	8.73	3.86	0.00	39.51	1.39 (1.47)	−1.2
6 months	8.67	3.83	0.00	49.28	1.78 (0.85)	−0.2
12 months	8.86	3.68	0.00	71.21	1.04 (0.88)	−3.0

- Source: text of the accepted version (26 August 2021); numbers from the authors' replication package of the published tables ([Zenodo, 2022](#))
- Valid: all 10 sample moments are positive, so $D_1 = 0$ and $p = 100\%$ at every horizon (Table 3)
 - ▶ Interpretation: no state of the nine predictors in 1990–2020 points to a violated bound
- Not tight: D_2 exceeds 27.13, the upper critical value of the 0.1% Kodde–Palm test, at every horizon; mean slackness is 4.8 to 5.2 points a year (Tables 1, 3)
 - ▶ Interpretation: the bound is less than half of the realised premium; validity comes from this average slackness, not from predictive power (Section 4.3)
- Forecasting: the slopes $\hat{\beta}$ are positive but imprecise (Table 5); with the nine predictors as controls, a one-standard-deviation rise in the 12-month bound adds 6.6 points to the 12-month excess return, significant at 1% at every horizon (Table 6)
 - ▶ Interpretation: the bound carries information, and the significant predictors show that slackness is predictable, so the bound is not tight
- Out of sample: the Martin bound as a forecast loses to the historical mean at every horizon; the best case, the Chabi-Yo–Loudis bound at 6 months, reaches 4.3%, not significant (Table 8)
 - ▶ Interpretation: 30 years of option prices are too few for a significant out-of-sample gain

Case Study: Replication, Limitations and an Open Extension

- **Replicate** (seminar C3): the authors' daily bound series ([replication package](#)), S&P 500 closes (data/market), the daily risk-free rate ([Kenneth French Data Library](#)), the Welch–Goyal predictors ([Amit Goyal's data page](#))
 - ▶ Rebuild Table 1 and the statistics D_1 , D_2 of Table 3; the course index has no dividends, so excess returns and slackness are lower: report it as a data limitation
- **Limitations stated by the authors**
 - ▶ A short series, limited to the years after 1990 by the need for option prices: the out-of-sample tests have little power (Section 6)
 - ▶ Validity holds even when the bound is replaced by its time-series mean, so it says nothing about predictive power (Section 4.3)
 - ▶ More recent bounds and formulas for risk premia are not tested (Introduction)
- **Open extension**: is the Martin bound still valid on 2021–2026 (the 2022 bear market, the 0DTE era of Section 9)? The course data have VIX, not SVIX: VIX weights OTM options by $1/K^2$ and exceeds SVIX when the left tail is priced ([Martin \(2017\)](#), Result 3); is a VIX-based bound still valid, and can the Kodde–Palm test detect it if not?

An Open Question: Does the VRP Still Predict Returns?

- [Bollerslev, Tauchen & Zhou \(2009\)](#), Table 2: monthly data 1990–2007, $y_t^{(h)} = \frac{12}{h} \sum_{j=1}^h (r_{m,t+j} - r_{f,t+j})$ on $\text{VRP}_t = \text{VIX}_t^2/12 - \text{RV}_t$
 - ▶ RV_t : realised variance of month t ; at $h = 3$: slope 0.47, Hodrick $t = 2.86$, adjusted $R^2 = 6.82\%$, the peak of a hump-shaped horizon profile
- [Goyal, Welch & Zafirov \(2024\)](#): 29 equity-premium predictors proposed after 2008, re-examined with data to 2021: over one third lose in-sample significance, half of the rest fail out of sample
- Open: is VRP predictability a stable price of variance and jump risk, or a property of 1990–2007? The 2008–2026 sample adds two crashes and the 0DTE era
- Formally: $H_0 : \beta_3 = 0$ vs $H_1 : \beta_3 > 0$ in $y_t^{(3)} = \alpha + \beta_3 \text{VRP}_t + u_t$
 - ▶ Out of sample: H_0 : the VRP forecast does not lower the mean squared prediction error of the historical mean; Clark–West statistic [Clark & West \(2007\)](#)

The Discovery Loop, Step by Step (1/2)

- **Literature:** "List peer-reviewed tests of VRP return predictability since 2009: DOI, sample, horizon, table, headline estimate."
 - ▶ Resolve every DOI on Crossref and read the cited table yourself; search with Semantic Scholar, Google Scholar, Elicit
- **Hypotheses:** "Give three competing explanations of VRP predictability and the testable implication of each for the horizon profile, for 2008-2026 and for non-US markets."
 - ▶ Time-varying risk aversion, jump-risk compensation, or a small-sample artefact of persistent regressors and overlapping returns
- **Data and code:** "Write Python that builds VIX squared/12 and monthly realised variance from data/market/VIX.INDX.csv and GSPC.INDX.csv and reproduces Table 2 of BTZ (2009), $h = 3$, with Hodrick (1992) standard errors."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); replicate the published number before extending anything

The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed before the results: "Draft a pre-analysis plan: horizons, out-of-sample start date, benchmark, test statistic and the multiple-testing correction across horizons."
 - ▶ Placebo: VRP lagged 12 months or block-shuffled; subsamples; 5-minute realised variance of SPY (from 2020) instead of daily squared returns
- **Critique**: "Act as a hostile referee: why could this out-of-sample R^2 be spurious or biased, and which test settles each objection?"
 - ▶ Expect: bias with persistent regressors, overlap, the choice of start date, the weight of 2008 and 2020
- **Report**: every horizon and specification tried, in one table; the plan dated before the results; AI use logged

What the Researcher Must Check

- **Invented or misquoted numbers:** an assistant "recalls" Table 2 of [Bollerslev, Tauchen & Zhou \(2009\)](#)
 - ▶ Test: compare with the paper itself, digit by digit; log any mismatch
- **Timing:** VIX at the end of month t , returns from $t + 1$, T-bill of the same month
 - ▶ Test: recompute the forecast for t on data truncated at t ; it must not change
- **Overlap:** OLS or Newey–West with few lags overstate t for 3-month returns sampled monthly
 - ▶ Test: Hodrick (1992) standard errors [Hodrick \(1992\)](#), or non-overlapping quarterly regressions
- **Horizon and start-date mining:** the best of eight horizons, the most favourable split
 - ▶ Test: report all horizons with a Holm correction; plot R_{OS}^2 against the start date
- **Measurement:** daily squared returns are not 5-minute realised variance
 - ▶ Test: match the published sample means (implied 33.23, realised 14.93) before any regression

Mini-Case: Replicate, Then Extend

- A typical assistant proposal: "the VRP predicts the quarterly equity premium; extend the 2009 result to today". We test it on the course data
- Replication, February 1990 to September 2007, $h = 3$, daily data: $\hat{\beta}_3 = 0.42$, $t_{\text{Hodrick}} = 2.25$, adjusted $R^2 = 4.6\%$ (published: 0.47, 2.86, 6.82%)
 - ▶ Mean implied variance 33.1 is close to 33.23; mean realised 20.6 vs 14.93: daily close-to-close data, a stated limitation
- Extension to May 2026 (436 months): $\hat{\beta}_3 = 0.09$, $t_{\text{Hodrick}} = 1.33$, adjusted $R^2 = 1.0\%$
- Out of sample from January 2008, 221 expanding-window forecasts: $R_{OS}^2 = 1 - \sum_t (y_t - \hat{y}_t)^2 / \sum_t (y_t - \bar{y}_t)^2 = -22.2\%$ [Campbell & Thompson \(2008\)](#); Clark–West $t = 0.64$ ($p = 0.26$): H_0 not rejected
- Reading: sign and size replicate in-sample; the longer sample does not support stable predictability. Outliers, measurement or a vanished premium: that is the open part

 [ai_discovery_case.py](#)

Your Turn: a Project Seed

- Question: does the VRP predict the equity premium after 2008, and does the Bitcoin premium of Section 7 (DVOL) predict Bitcoin returns?
- Replicate first: [Bollerslev, Tauchen & Zhou \(2009\)](#), Table 2, $h = 3$: slope 0.47, $t = 2.86$, adjusted $R^2 = 6.82\%$; explain every gap
- Deliverables
 - ▶ A dated pre-analysis plan; the replication table; out-of-sample results for all horizons with a Holm correction
 - ▶ Robustness (5-minute variance, subsamples, without 2008 and 2020); a referee-style critique and your reply
- Log each prompt that shaped a decision in `AI_USE.md`; each invented reference, wrong number or look-ahead caught in `AI_ERRORS.md`, with the test that caught it ([templates](#))

What We Learned

- ▣ Parity, bounds and static spanning need no model; the tree and the Black–Scholes equation price by dynamic replication
- ▣ The Greeks measure exposures; discrete hedging leaves an error of order $1/\sqrt{N}$
- ▣ A delta-hedged option is a bet on realised vs implied variance; implied variance exceeded realised variance on average on the S&P 500 and on Bitcoin
- ▣ The surface and the risk-neutral density are estimates: impose no-arbitrage constraints and report bands
- ▣ The VIX is a strip estimator with jump and truncation biases; as a forecast it needs joint tests, logs and HAC loss comparisons

Frontier: Research on Option-Implied Information

- Andersen, Fusari & Todorov (2015): a left-tail factor in index-option surfaces, not spanned by volatility, predicts future equity and variance premia
- Martin (2017): under the negative correlation condition, $R_f \text{SVIX}^2$ bounds the equity premium from below; the bound exceeded 20% at the height of the 2008 crisis
- Bali et al. (2023): nonlinear machine learning on option and stock characteristics predicts equity-option returns out of sample (Chapter 13)
- Open: jointly estimated, arbitrage-free surfaces with valid confidence bands; the pricing of 0DTE options

Coming Next and Further Reading

- Chapter 13: machine learning for financial prediction, with volatility and option-implied features among the inputs
- Options and the implied-volatility surface: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017), Čížek, Härdle & Weron (2011), Gatheral (2006)
- Pricing kernels from option prices: Aït-Sahalia & Lo (1998), Grith, Härdle & Park (2013)
- Crypto options and their risk: Matic, Packham & Härdle (2023), Alexander & Imeraj (2020), Pele et al. (2023)

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  XFG Advanced Methods in Quantitative Finance
- ▣  Pricing Kernels and Risk Premia
- ▣  Hedging Cryptocurrency Options

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_12

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